

Crystallization Effects on Space Charge Accumulation in Polypropylene under DC Voltage

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Abstract- Polypropylene (PP) is a potential recyclable solid dielectric with promising applications in high voltage engineering. It is known that the crystallization of polypropylene influences its mechanical properties. However, there is no clear correspondence between morphological features and space charge accumulation in PP. In the present work, the polypropylene material was thermally treated at 110°C to re-crystallize the sample. Using the principle of the pulse electroacoustic (PEA) method, the effect of re-crystallization on charge accumulation was investigated. It has been found that while re-crystallization occurs, the mesophase and charge traps in PP crystallization increase, leading to the charge migration rate and the total charge accumulation increasing.

I. INTRODUCTION

As the insulating material of mainstream high voltage direct current (HVDC) cables, cross-linked polyethylene (XLPE) meets the mechanical and thermal properties needs of HVDC extruded cable insulation, but it also has some limitations. The cross-linking process of XLPE consumes a lot of energy and has low production efficiency, for it needs high pressure and high temperature, precise temperature control, and degassing to reduce the influence of cross-linked by-products on DC conductivity and space charge accumulation [1]. In addition, the cross-linked by-products will inevitably lead to accelerated aging of the cable insulation, making it difficult for XLPE cables to reach the designed service life of 40 years [2]. Finally, XLPE is not a sustainable material, and its recycling is difficult and has a great adverse impact on the environment [3]. Given the above problems, XLPE has been unable to meet the needs of today's ecological protection and social and economic development.

Compared to XLPE, polypropylene (PP) has the advantages of higher operating temperature, being fully recyclable and cross-linking by-product free. Therefore, PP is expected to become the sustainable insulating material for HVDC cables. PP is a typical semi-crystalline material, and its morphological features can be influenced by its thermal history. Thermal treatment refers to the process of placing materials in a certain medium for heating, heat preservation and cooling. For semi-crystalline polymers, thermal treatment at an appropriate temperature leads to re-crystallization inside semi-crystalline polymers, transforming a defective crystallization into a perfect one [4], the transformation of tiny crystal grains into larger crystal grains [5], and the thickening of thin initial

lamella [6]. The actual operating temperature of the cable can reach the temperature at which the PP re-crystallizes.

One of the key issues for HVDC insulation is space charge accumulation which may affect charge transport and electric field distribution. A similar concern occurs for PP application for HVDC cables. When the voltage is higher than the range of the common engineering field (20kV/mm), the formation of space charge has been associated with aging, breakdown, etc [7]. Space charge can cause local electric field distortion and electrical performance degradation in PP cable insulation, reducing the service life of the cable. The space charge measurement can help us better study the electrical properties of PP. Currently, the principle of the pulse electroacoustic (PEA) method is widely used in space charge measurement [8].

Studies have also found that the semi-crystalline material has a possible incidence of larger trap density [9], which is influenced by the micromorphology of semi-crystalline. In summary, the PP re-crystallization temperature has been reached in the actual operation of HVDC cables, and the PP micromorphology may impact its space charge accumulation. Therefore, studying the effect of re-crystallization on the accumulation of space charges has a certain guiding significance for the actual operation of HVDC cables. In this paper, PP film samples were thermally treated at a low temperature (110 °C, at which re-crystallization can occur) for a variety of times – the melting temperature of PP is about 160 °C. Space charge dynamics and thermodynamics were measured to explore the effects of crystallization on space charge accumulation in PP.

II. EXPERIMENTAL METHODS

A. Sample Preparation

The polypropylene (PP, isotactic) used in this study is purchased from the supplier Sigma Aldrich. Its weight average molecular weight (Mw) is approximately 340,000, and its number average molecular weight (Mn) is approximately 97,000. The original form of isotactic polypropylene (iPP) is pelleting with a relative density of 0.9 g/mL at 25 °C. Before thermal treatment, all PP materials were pressed to the size required by a hydraulic press at 180 °C and quenched in the ionized water at room temperature of 25 °C to get relatively simple morphology [10]. The thermal treatment consisted of successive annealing at 110 °C in the air for a total time ranging between 0 minutes and 30 minutes.

B. Polarized Light Microscopy (PLM)

It is possible to use PLM to obtain the crystal morphology of PP macroscopically. This PLM is an optical microscopy technique involving polarized light. A simple technique involves irradiating the sample with polarized light. Choose to use a polarizer at 90° for the illumination to block directly transmitted light. Birefringence is the basic feature of crystals, and materials with birefringence are studied and identified by POM using the polarization characteristics of light. PLM can be used to observe the crystallization of PP.

C. Differential Scanning Calorimetry (DSC)

The melting process of the samples with different thermal treatment times was characterized by DSC. The sample (about 5 mg) was heated from 20 °C to 200 °C at 10 °C/min rate. The degree of crystallinity (X_c) of PP can be calculated by the melting enthalpies according to [8]:

$$X_c = \frac{\Delta H_m}{\Delta H_m^\infty} \quad (1)$$

where ΔH_m is the measured melting enthalpy of the sample and ΔH_m^∞ refers to the melting enthalpy for 100 % crystalline iPP (50 Cal/g, 209 J/g [11]).

D. Space Charge Measurement

The principle of the pulse electroacoustic (PEA) was used to identify space charge distribution and obtain charge accumulation within the sample. The PEA system consists of an anode made of a semiconducting composite material (semicon) and a cathode made of aluminum. A drop of silicone oil was used to ensure good contact between the sample and the electrodes for the transmission of acoustic signals. Acoustic signals of charge in samples were collected for 60 minutes under an applied electric field of 40 kV/mm and short-circuited in succession.

In order to eliminate the influence of the capacitive charge on the electrodes on the data, a subtraction method has been used to determine the threshold field of space charge injection in the dielectric. Space charge density after subtraction $\rho_{acc}(x, t)$ can be obtained with equation (2) below [12],

$$\rho_{acc}(x, t) = \rho_{app}(x, t) - \frac{V_{appl}}{V_{ref}} \rho_{ref}(x, t) \quad (2)$$

where x is the distance to the cathode; t is the field application time; reference data is collected under a low electric field; $\rho_{ref}(x, t)$ and $\rho_{app}(x, t)$ represents the charge density of reference data and experimentally measured data at point x and time t , respectively; V_{ref} and V_{appl} are the reference and applied electric field, respectively [12].

Moreover, the total amount of charge can be calculated with equation (3) below [13][14],

$$Q = \int_0^d |\rho(x)| \cdot A \cdot dx \quad (3)$$

where Q is the total charge amount, $\rho(x)$ is charge density, A is the electrode area (50.265 mm²) and d is the thickness of the sample.

III. RESULTS AND ANALYSIS

A. Polarized Light Microscopy(PLM)

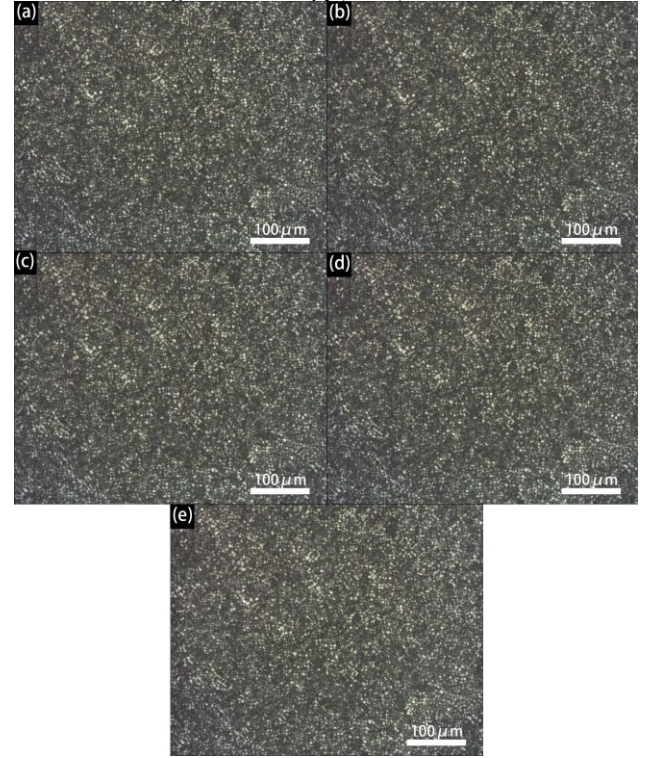


Fig. 1. PLM images of PP with thermal treatment time of (a) 0 min, (b) 5min, (c) 15min, (d) 20min, and (e) 30min, respectively.

The quenched PP sample can be seen in the PLM with a 500 times magnified image (Fig. 1 (a)), and the crystals show a feature of small diameter and uniformity. The small diameter of the PP crystals in the image indicates that the quenched PP sample has no time for crystal growth. After thermal treatment at 110°C at different times, the macroscopic morphology of the PP sample crystals did not change significantly, and there was no apparent new crystal formation.

B. Differential Scanning Calorimetry (DSC)

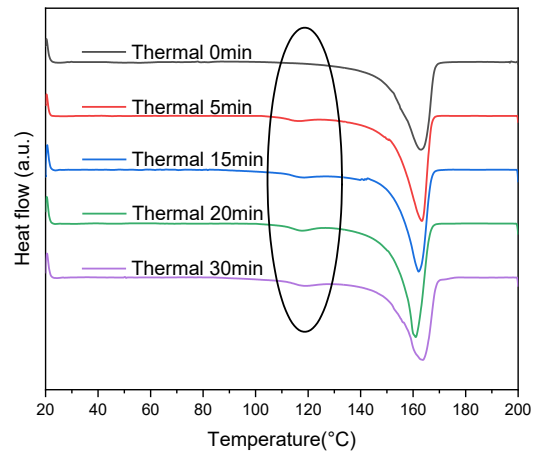


Fig. 2. Melting curves of PP with 110°C thermal treatment for different times

Melting curves can reflect the changes in crystallinity of PP samples with the different thermal treatment times. Figure 2 shows the melting curves of PP samples with thermal treatment consisting of successive annealing at 110 °C. The melting peak occurs at about 160 °C. At about 115 °C, a new endothermic peak (black oval) can be seen after thermal treatment. With the successive annealing time increase, the endothermic peak becomes more significant. The new endothermic shows thermal treatment leads to re-crystallization in the PP samples.

As shown in Table I, the crystallinity of all PP samples is lower than 7.5 % because the quenching pretreatment of the PP samples makes the PP crystallization time very short, and the PP crystals cannot be formed and grown. From 0 to 20 minutes, with the increase in successive annealing time, the crystallinity of the sample has a slight increase of about 0.5 %. From 20 to 30 minutes, the crystallinity of PP samples does not change. This shows that in the thermal treatment at 110 °C, the re-crystallization of PP completes in 20 minutes. After 20 minutes, PP samples become thermally stable at 110 °C.

TABLE I

MELTING PROPERTIES OF PP WITH DIFFERENT THERMAL TREATMENT TIME					
Sample thermal time(min)	0	5	15	20	30
Melting peak temperature(°C)	163	163	162	161	163
Melting enthalpy(J/g)	14.1	14.5	14.7	15.3	15.2
Crystallinity(%)	6.75	6.93	7.03	7.32	7.27

C. Space Charge Measurement

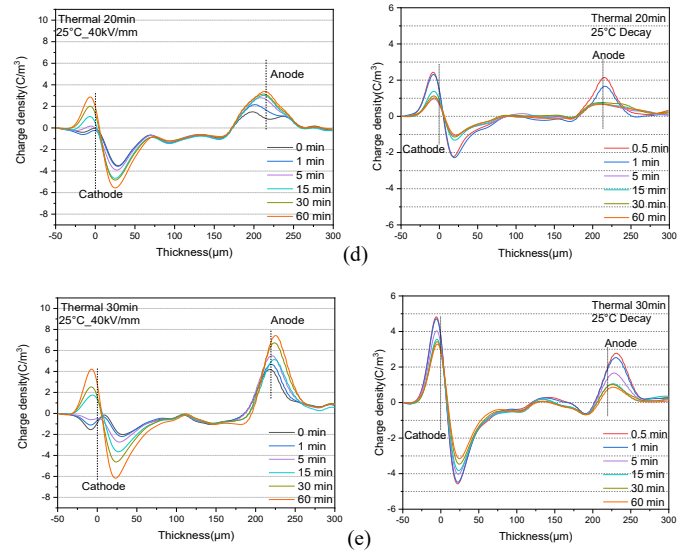
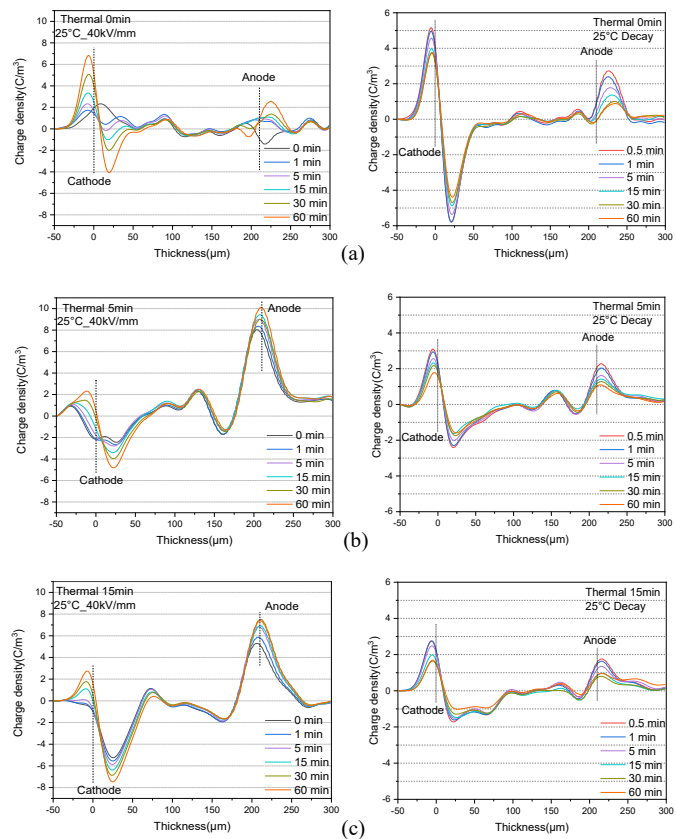


Fig. 3. Space charge distribution (left) and space charge decay (right) profiles of PP samples with thermal treatment time of (a) 0min, (b) 5min, (c) 15min, (d) 20min and (e) 30min at DC 40 kV/mm in 60 min

Space charge accumulation is a concerning problem in HVDC cables, which can induce electric field distortion and shorten insulation lifetime. The re-crystallization inside PP changes the crystallization microscopic morphology and may have some impact on the accumulation of space charges. The results in Figure 3 were obtained after applying equation (2), which shows space charge distribution and space charge decay profiles in samples with different thermal treatment times at an applied electric field of 40kV/mm over 60 minutes in succession.

Compared with the untreated sample, the space charges in the thermally treated sample migrate rapidly, and the injection depth also increases. At the thickness of 100 μm, with the thermal time increase, the charge distribution tends to be flatter and close but lower than 0 C/m³. After thermal treatment, the increase in the migration speed of negative charges is greater than that of positive charges. As it can be seen in the space charge decay profiles, the space charge decay in the sample as the depolarization time becomes longer. More information on the total charge accumulation can be obtained from Figure 4. Samples with thermal treatment generally have a higher charge injection than the untreated samples. The decay rate for all the samples does not show a significant difference.

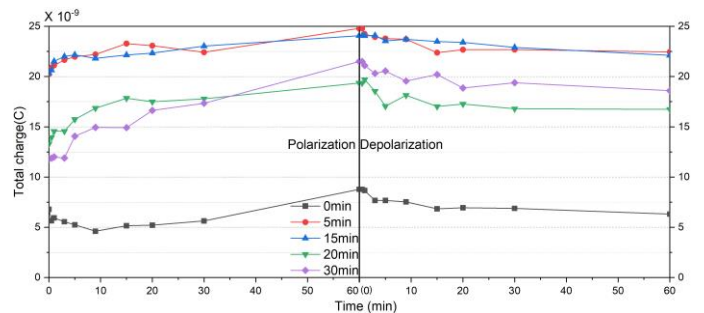


Fig. 4. Total space charge accumulation of PP samples with different thermal treatment time during polarization and depolarization

Combining re-crystallization phenomena shown in DSC results, the increase of charge injection and charge migration speed inside thermally treated PP sample shown in space charge dynamics results can be explained. Thermal treatment favors molecular mobility in the amorphous phases [5]. The amorphous chains in amorphous phases move close to crystal organization to achieve a more stable arrangement. Before the amorphous chains move to a stable thermodynamic position, it is in the mesophase between the crystal lamellae and the amorphous phase[15].

This mesophase, right next to crystal lamellae, increases the surface traps of PP crystallization. Driven by the electric field, charges are easily trapped and detrapped by traps, which increases charge injection and charge migration inside samples [13]. During the re-crystallization (thermal treatment for 5 minutes and 15 minutes), the increase of the mesophase caused a large number of traps on the crystallization surface so that the total charge injection amount reached 20×10^{-9} C at the moment of voltage application, which was three times that of the un-thermally treated sample (6.79×10^{-9} C). When the applied electric field is removed during the depolarization, the total charge in the thermally treated samples is significantly greater than that in the untreated samples. This phenomenon indicates that the traps in samples with thermal treatment are increased. When the re-crystallization reaches a thermally stable state, the mesophase is transformed into the crystal lamellae, the mesophase is reduced, the traps on the crystallization surface are reduced, and the amount of charge injection decreases to a certain extent. However, the total amount of charge injection is still higher than that of the un-thermally treated sample.

IV. CONCLUSIONS

The thermal treatment at 110 °C causes re-crystallization inside the quenched PP sample, and the sample reaches a relatively thermally stable state after 20 minutes of thermal treatment.

The increase in mesophase during re-crystallization leads to traps increase, resulting in increased charge injection and charge migration inside PP.

With the completion of re-crystallization, the thermal state of PP tends to be stable, and the amount of charge injection will decrease, but it is still higher than that of the quenched sample. The re-crystallized internal structure - mesophase, is favorable for the charge migration when an electric field is applied because of increasing traps. Thermal treatment favors molecular mobility in the amorphous phases. The amorphous chains in amorphous phases move close to crystal organization to achieve a more stable arrangement. Before the amorphous chains move to a stable thermodynamic position, it is in the mesophase between the crystal lamellae and the amorphous phase.

Future work will observe the re-crystallization phenomenon at higher temperatures to obtain more obvious morphology changes and explore the effects of PP morphology and space charge traps.

REFERENCES

- [1] T. Andritsch, A. Vaughan, and G. C. Stevens, "Novel insulation materials for high voltage cable systems," *IEEE Electr. Insul. Mag.*, vol. 33, no. 4, pp. 27–33, Jul. 2017.
- [2] G. Mazzanti, "Life and reliability models for high voltage DC extruded cables," *IEEE Electr. Insul. Mag.*, vol. 33, no. 4, pp. 42–52, Jul. 2017.
- [3] J. He, L. Peng, and Y. Zhou, "Research progress of environment-friendly HVDC power cable insulation materials," *Gaodianya Jishu/High Volt. Eng.*, vol. 43, no. 2, pp. 337–343, 2017.
- [4] R. S. Schotland, "Influence of annealing on the density and tensile properties of polypropylene," *Polym. Eng. Sci.*, vol. 6, no. 3, pp. 244–247, Jul. 1966.
- [5] L. Poussin, Y. A. Bertin, J. Parisot, and C. Brassy, "Influence of thermal treatment on the structure of an isotactic polypropylene," *Polymer (Guildf.)*, vol. 39, no. 18, pp. 4261–4265, Jan. 1998.
- [6] S. Rastogi, A. B. Spoelstra, J. G. P. Goossens, and P. J. Lemstra, "Chain Mobility in Polymer Systems: on the Borderline between Solid and Melt. 1. Lamellar Doubling during Annealing of Polyethylene," *Macromolecules*, vol. 30, no. 25, pp. 7880–7889, Dec. 1997.
- [7] S. Boggs, "A rational consideration of space charge," *IEEE Electr. Insul. Mag.*, vol. 20, no. 4, pp. 22–27, 2004.
- [8] Y. Zhou, N. Wang, Y. Wang, Q. Sun, X. Liang, and Z. Guan, "Review of research on space charge in solid dielectrics," *Diangong Jishu Xuebao/Transactions China Electrotech. Soc.*, vol. 23, no. 9, pp. 16–25, 2008.
- [9] S. Das and N. Gupta, "Charge trapping and transport phenomenon in aged and unaged epoxy resin and polyethylene," *Int. Trans. Electr. Energy Syst.*, vol. 25, no. 6, pp. 948–957, Jun. 2015.
- [10] I. L. Hosier, A. S. Vaughan, and S. G. Swingle, "An investigation of the potential of ethylene vinyl acetate/polyethylene blends for use in recyclable high voltage cable insulation systems," *J. Mater. Sci.*, vol. 45, no. 10, pp. 2747–2759, 2010.
- [11] J. R. Isasi, L. Mandelkern, M. J. Galante, and R. G. Alamo, "The Degree of Crystallinity of Monoclinic Isotactic Poly(propylene)," *J. Polym. Sci. Part B Polym. Phys.*, vol. 37, no. 4, pp. 323–334, 1999.
- [12] N. Liu, C. Zhou, G. Chen, and L. Zhong, "Determination of threshold electric field for charge injection in polymeric materials," *Appl. Phys. Lett.*, vol. 106, no. 19, p. 192901, May 2015.
- [13] G. Chen and Z. Xu, "Charge trapping and detrapping in polymeric materials," *J. Appl. Phys.*, vol. 106, no. 12, 2009.
- [14] T. Zhou, G. Chen, R. Liao, and Z. Xu, "Charge trapping and detrapping in polymeric materials: Trapping parameters," *J. Appl. Phys.*, vol. 110, no. 4, p. 043724, Aug. 2011.
- [15] S. Gautam, S. Balijepalli, and G. C. Rutledge, "Molecular Simulations of the Interlamellar Phase in Polymers: Effect of Chain Tilt," *Macromolecules*, vol. 33, no. 24, pp. 9136–9145, Nov. 2000.